

Two-Photon Physics at the $LH\mu C$

High-energy $\gamma\gamma$ interactions in μp collisions

Hamzeh Khanpour
AGH University of Krakow

(on behalf of HELP4CERN Study Group)

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- 5 Heavy and BSM final states
- 6 Experimental opportunities and outlook

Why a muon-hadron collider for two-photon physics?



- A muon-hadron collider combines a clean lepton-hadron initial state with a multi-TeV centre-of-mass energy.
- Replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, making much higher lepton-beam energies possible.
- The μp environment remains experimentally cleaner than pp for many exclusive and semi-exclusive final states: negligible pile-up is a central advantage of the lepton-hadron concept.
- Both beams radiate photons. Their electromagnetic fields therefore create a high-energy $\gamma\gamma$ collider *inside* the μp machine.

A high-energy $\gamma\gamma$ collider!

LH μ C combines a high-energy photon source from the muon beam with the electromagnetic photon field of the proton, opening a distinctive high-energy $\gamma\gamma$ programme.

LH μ C@CERN: benchmark working points



- The present talk focuses on the two-photon physics case of the LH μ C, using the current working benchmarks:
- The first option: $E_\mu = 500$ GeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 3.7$ TeV, with a $\mathcal{L}_{\text{int}} = 250$ fb $^{-1}$.
- **The nominal baseline:** $E_\mu = 1$ TeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 5.3$ TeV, with a $\mathcal{L}_{\text{int}} = 1000$ fb $^{-1}$.
- The second option: $E_\mu = 2$ TeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 7.5$ TeV, with a $\mathcal{L}_{\text{int}} = 1000$ fb $^{-1}$.

LH μ C physics message

These working points are sufficient to demonstrate the main message of LH μ C: already at the 500 GeV, LH μ C enters the multi-TeV regime and becomes highly competitive for $\gamma\gamma$ physics, Higgs, top, electroweak, and representative BSM studies.

Benchmark WPs

Machine	Benchmark
LHeC	$E_e = 20, 50$ GeV
LH μ C (option 1)	$E_\mu = 500$ GeV
LH μ C (baseline)	$E_\mu = 1000$ GeV
LH μ C (option 2)	$E_\mu = 2000$ GeV
Configuration	Value
$\sqrt{s}_{\mu p}$ (option 1)	$\simeq 3.7$ TeV
$\sqrt{s}_{\mu p}$ (baseline)	$\simeq 5.3$ TeV
$\sqrt{s}_{\mu p}$ (option 2)	$\simeq 7.5$ TeV
$\mathcal{L}_{\text{int}}$ (option 1)	250 fb $^{-1}$
$\mathcal{L}_{\text{int}}$ (baseline)	1000 fb $^{-1}$
$\mathcal{L}_{\text{int}}$ (option 2)	1000 fb $^{-1}$

Representative benchmark cross sections



Why is this relevant for $\gamma\gamma$ physics?

Moving from the LHeC to the LH μ C strongly increases the available lepton-hadron centre-of-mass energy and opens the multi-TeV regime. The same energy lever arm is crucial for two-photon physics: the photon spectrum from the high-energy muon beam becomes significantly harder, extending the accessible $W_{\gamma\gamma}$ range. This enhances the reach for heavy $\gamma\gamma$ final states and increases sensitivity to new interactions in the high-mass tails.

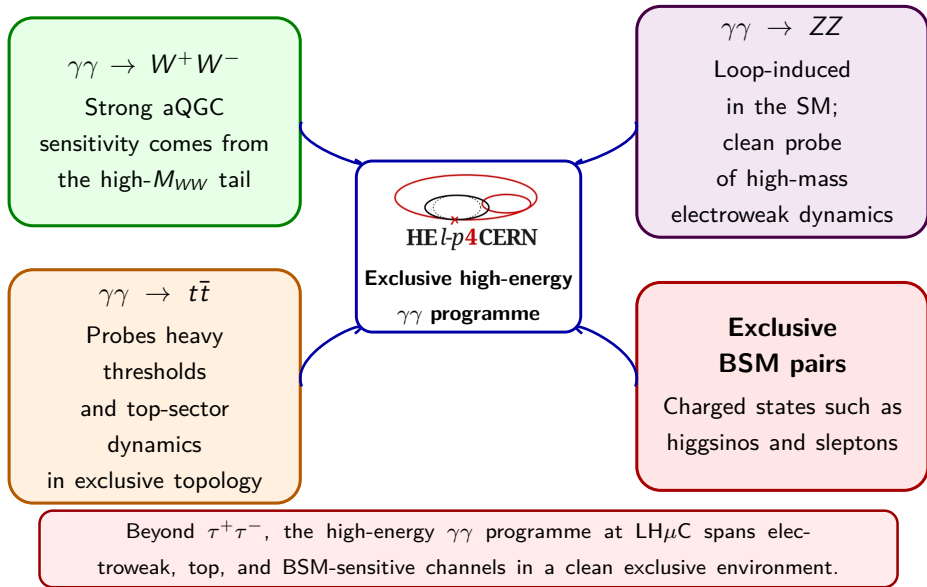
Benchmark inclusive cross sections

Process	20 GeV	50 GeV	500 GeV	1 TeV
	LHeC	LHeC	LH μ C	LH μ C
$\sigma(\ell p \rightarrow \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \rightarrow \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \rightarrow \ell t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \rightarrow \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

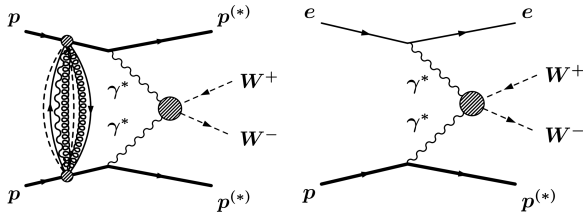
Key comparisons

Relative to the 50 GeV LHeC benchmark, the nominal 1 TeV LH μ C baseline enhances the benchmark inclusive rates by approximately: ~ 4.5 for charged-current Higgs production, ~ 12 for neutral-current Higgs production, ~ 13 for single-top production, and ~ 69 for $t\bar{t}$ production.

Exclusive $\gamma\gamma$ channels: electroweak, top, and BSM reach



Two-photon production in lepton-proton (ℓp) scattering



pp: two hadronic beams

ℓp : one hadronic beam

Two-photon searches at the LHC already demonstrate the physics potential of exclusive and semi-exclusive $\gamma\gamma$ interactions. Moving to a lepton-hadron collider removes one of the two hadronic initial states and therefore considerably simplifies the event environment.

Factorised $\gamma\gamma$ picture

$$\ell \rightarrow \ell + \gamma^*, \quad p \rightarrow p^{(*)} + \gamma^*,$$

$$\gamma\gamma \rightarrow X,$$

$$\boxed{\ell p \rightarrow \ell + X + p^{(*)}}$$

The incoming lepton and proton act as sources of quasi-real photons. The hard central system X is then produced through $\gamma\gamma$ fusion.

From LHeC to LH μ C

At the LH μ C, the higher lepton-beam energy produces a harder equivalent-photon spectrum, extending the accessible $W_{\gamma\gamma}$ range and strengthening the reach for heavy final states and high-mass EFT/BSM effects.

$\gamma\gamma$ at ℓp : production topologies and experimental handles



Photon-emission topologies

• Elastic

$$p \rightarrow p + \gamma^*$$

The proton remains intact. Its electromagnetic form factors $F_E(Q^2)$ and $F_M(Q^2)$ determine the elastic photon flux.

• Proton-dissociative

$$p \rightarrow X_p + \gamma^*, \quad M_N > m_p.$$

The proton breaks up into a hadronic system X_p . The photon flux is then described using proton structure functions. Allowing larger proton-dissociation masses M_N and photon virtualities Q_p^2 increases the accessible $\gamma\gamma$ luminosity.

Experimental handles

- **Low pile-up** preserves clean final states and makes rapidity-gap / exclusivity requirements particularly effective.
- **Reduced hadronic re-scattering** compared with pp : there is no second hadronic beam and hence no analogous soft-survival complication on the lepton side.
- The **scattered lepton** constrains the photon emission from the lepton side; radiative corrections are predominantly QED, including ISR.
- A **forward proton tag**, when available, can over-constrain the event kinematics and is especially valuable for final states containing missing energy.

Equivalent Photon Approximation: factorising the ℓp cross section



EPA master relation

$$\sigma_{\mu p} = \int dy_\mu dy_p \Phi_\mu(y_\mu) \Phi_p(y_p) \sigma_{\gamma\gamma}(W) = \int dW S_{\gamma\gamma}(W) \sigma_{\gamma\gamma}(W)$$

- Φ_μ and Φ_p are the equivalent-photon fluxes radiated by the two beams.
- The hard subprocess $\gamma\gamma \rightarrow X$ factorises from photon emission, in close analogy with partonic convolutions.
- $S_{\gamma\gamma}(W)$ packages the two fluxes into the effective two-photon luminosity spectrum.

Kinematic variables

$$y_\mu = \frac{E_\gamma^{(\mu)}}{E_\mu}, \quad y_p = \frac{E_\gamma^{(p)}}{E_p},$$

$$W \equiv W_{\gamma\gamma}.$$

Equivalent photon fluxes from the leptons (e or μ^+) and proton



Generic EPA flux

$$\Phi_\gamma(y) = \frac{\alpha}{\pi y} \int \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right]$$

$$Q_{\min}^2 = \frac{M^2 y^2}{1-y}.$$

- Point-like lepton: $F_E = F_M = 1$ and $M = m_\mu$.
- The muon mass raises $Q_{\min}^2$ relative to an electron at the same photon energy fraction y .

Proton dissociation

For $p \rightarrow X$, structure functions replace elastic form factors and the flux depends on the dissociative mass M_N .

Elastic proton emission

$$F_M = G_M^2,$$

$$F_E = \frac{4m_p^2 G_E^2 + Q^2 G_M^2}{4m_p^2 + Q^2}.$$

The dipole form factors suppress large Q_p^2 .

Beyond the simplest EPA: photon virtuality and exact $W_{\gamma\gamma}$ kinematics

High-energy approximation

$$W^2 \simeq y_\ell y_p s.$$

Exact elastic relation in the updated study

$$W^2 = -Q_\ell^2 + y_\ell y_p s$$

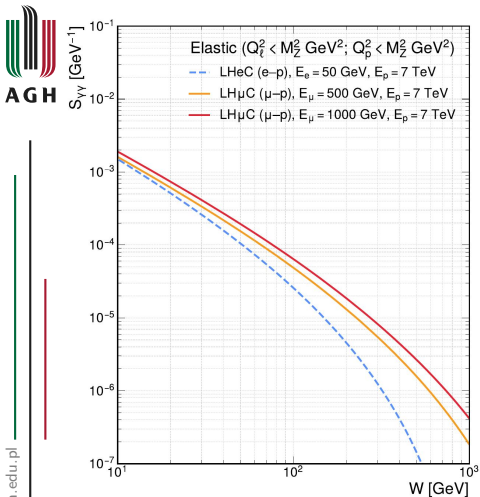
The paper writes $\ell = e$; the same lepton-side kinematics applies after $e \rightarrow \mu$.

- The approximation is excellent when photon virtualities are negligible relative to W^2 .
- For proton dissociation, the exact expression also depends on Q_p^2 and M_N .
- Elastic EPA agrees at percent level with matrix-element generators once the proper W definition is used; inelastic modelling is more sensitive to the dissociative phase space.

Precision lesson

At sizeable Q^2 , correct $W_{\gamma\gamma}$ reconstruction is part of the physics model, not a cosmetic refinement.

What does the $\gamma\gamma$ luminosity spectrum $S_{\gamma\gamma}$ encode?



Elastic photon-photon luminosity spectrum $S_{\gamma\gamma}$ as a function of the $\gamma\gamma$ invariant mass W , comparing the LHeC and LH μ C benchmark configurations.

$S_{\gamma\gamma}$ tells us

$S_{\gamma\gamma}(W)$ is the effective photon-photon luminosity density obtained from the convolution of the equivalent-photon fluxes of the lepton and proton. It is the universal machine-level input that determines how much luminosity is available at a given $\gamma\gamma$ invariant mass W .

- Increasing the lepton-beam energy shifts the usable $\gamma\gamma$ luminosity to larger W : the LH μ C spectra fall more slowly than the LHeC spectrum in the high-mass region.
- Already for $E_\mu = 500$ GeV, the accessible high- W region is substantially enlarged; the 1 TeV nominal option extends it further.

From luminosity to a physical process

$$\sigma_{\ell p \rightarrow \ell X p^{(*)}} = \int dW S_{\gamma\gamma}(W) \sigma_{\gamma\gamma \rightarrow X}(W).$$

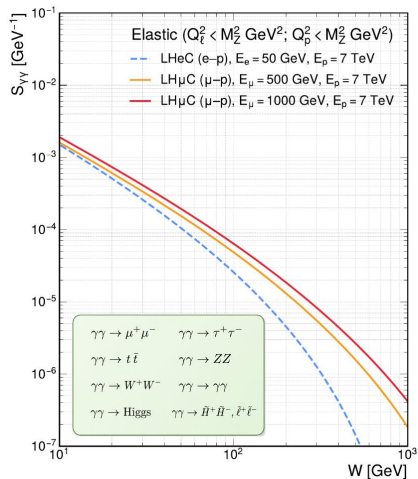
$S_{\gamma\gamma}$ is process independent: the final-state physics enters through the subprocess cross section $\sigma_{\gamma\gamma \rightarrow X}$.

High-energy $\gamma\gamma$ physics is a natural part of the LH μ C programme



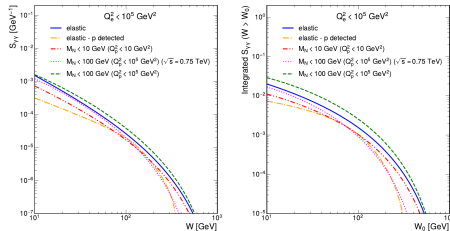
Interesting HET directions

- Once the photon luminosity is folded with the relevant $\gamma\gamma$ subprocess, the LH μ C provides access to a broad and complementary two-photon programme.
- Precision QED and lepton physics: $\gamma\gamma \rightarrow \mu^+\mu^-, \tau^+\tau^-$, including anomalous τ electromagnetic moments,
- Electroweak and EFT probes: $\gamma\gamma \rightarrow W^+W^-$ and sensitivity to anomalous quartic gauge couplings,
- Rare neutral-gauge channel: loop-induced $\gamma\gamma \rightarrow ZZ$, with sensitivity to neutral quartic gauge interactions,
- Heavy charged final states: $\gamma\gamma \rightarrow t\bar{t}$ and other high-threshold processes,
- BSM opportunities: light-by-light scattering and pair production of new electrically charged particles.



Elastic $S_{\gamma\gamma}(W)$ for the LHeC and LH μ C benchmark configurations.

Elastic vs. inelastic $\gamma\gamma$ luminosity and forward proton tagging



LHeC reference study

The detailed calculation separates elastic, tagged elastic ($0.01 < y_p < 0.1$), and proton-dissociative photon emission for several M_N and Q_p^2 ranges.

- Larger allowed dissociative masses increase the usable $\gamma\gamma$ luminosity.
- A forward proton tag improves kinematic closure, especially for final states with neutrinos.
- Low pile-up means proton tagging need not be mandatory for every clean channel.

From photon luminosity to physics: $\gamma\gamma \rightarrow X$



Collider rate

$$\sigma(\mu p \rightarrow \mu X p^{(*)}) = S_{\gamma\gamma} \otimes \sigma(\gamma\gamma \rightarrow X).$$

$\ell^+\ell^-$
calibration

$\tau^+\tau^-$
dipole
moments

W^+W^-
gauge
structure

ZZ
loop / nQGC

$t\bar{t}$
heavy
threshold

charged BSM
pairs

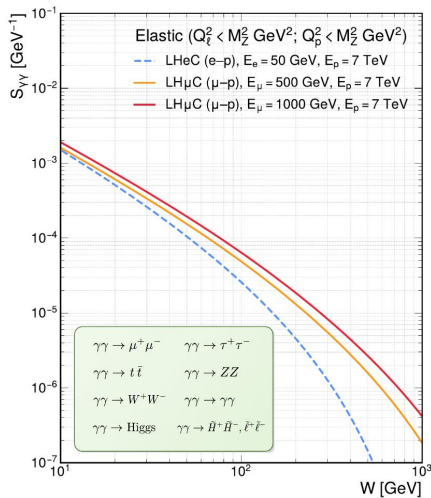
$\gamma\gamma$
light-by-light

H
loop-induced

Take-home message

The same photon luminosity supports precision QED, electroweak and loop physics, heavy thresholds, and direct searches for charged new states.

The high-energy $\gamma\gamma$ physics landscape at LH μ C



- $\ell\ell, \tau\tau$: high-rate QED calibration and dipole probes.
- WW : sizeable SM rate and direct $\gamma\gamma WW$ coupling.
- ZZ : loop-induced SM benchmark and neutral quartic couplings.
- $t\bar{t}$: heavy threshold and top electromagnetic structure.
- Charged BSM pairs: production controlled mainly by electric charge and mass.
- $\gamma\gamma$ and H : loop-induced quantum processes and complementary searches.

$\gamma\gamma \rightarrow \ell^+\ell^-$: a standard candle for two-photon physics

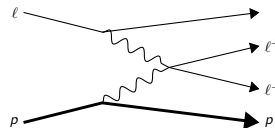


- e^+e^- and $\mu^+\mu^-$ provide clean control samples for the complete $\gamma\gamma$ programme.
- They test EPA modelling, acoplanarity and elastic/inelastic separation.
- At LHeC, elastic EPA predictions agree with GRAPE and CepGen/LPAIR at the percent level for matched cuts and exact W kinematics.

Elastic $\mu^+\mu^-$ benchmark at LHeC

$$Q_{e,p}^2 < 10 \text{ GeV}^2$$

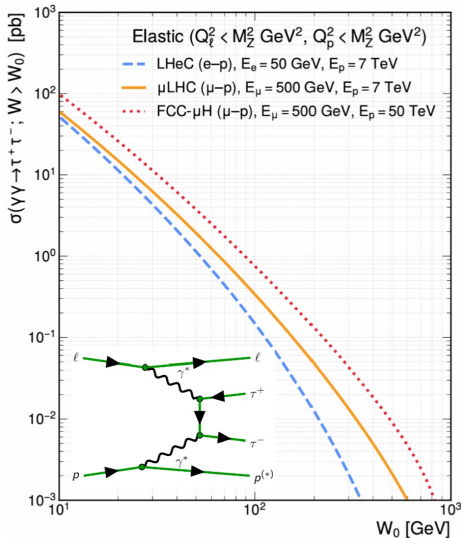
	$W_0 = 10 \text{ GeV}$	$W_0 = 100 \text{ GeV}$
EPA	120.12 pb	0.193 pb
CepGen/LPAIR	118.31 pb	0.191 pb
GRAPE (BH)	121.37 pb	0.193 pb



Why it matters for LH μ C

The same QED process becomes a high-statistics calibration candle for the harder μp photon spectrum and any forward-tagging strategy.

$\gamma\gamma \rightarrow \tau^+\tau^-$: a high-statistics tau factory



Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 50.37 \text{ pb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 58.56 \text{ pb}$$

- Large statistics make $\tau\tau$ a natural precision channel.
- LH μ C extends the accessible $M_{\tau\tau}$ tail.
- The hard tail enhances sensitivity to non-standard $\gamma\tau\tau$ interactions.

Anomalous τ electromagnetic moments: the $\gamma\tau\tau$ vertex

General electromagnetic vertex

$$\Gamma^\mu(q^2) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m_\tau} F_2(q^2) + \sigma^{\mu\nu}q_\nu \gamma_5 F_3(q^2),$$

$$F_2(0) = a_\tau, \quad F_3(0) = -\frac{2m_\tau}{e} d_\tau.$$

- a_τ : CP-even anomalous magnetic dipole moment.
- d_τ : CP-odd electric dipole moment.
- $\gamma\gamma \rightarrow \tau^+\tau^-$ probes the vertex directly.

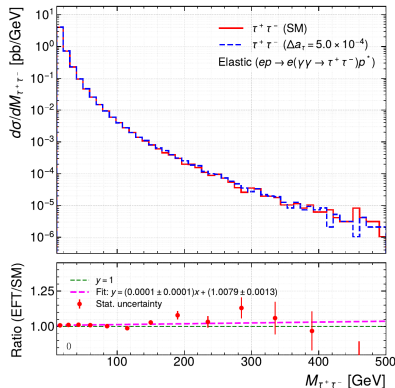
SMEFT interpretation

Gauge-invariant tau dipoles:

$$\bar{L}_\tau \sigma^{\mu\nu} \tau_R H \left(\frac{C_{\tau B}}{\Lambda^2} B_{\mu\nu} + \frac{C_{\tau W}}{\Lambda^2} W_{\mu\nu} \right) + \text{h.c.}$$

Real and imaginary coefficient combinations map respectively to magnetic and electric dipole deformations after electroweak symmetry breaking.

Probing τ dipole moments in the high- $M_{\tau\tau}$ tail



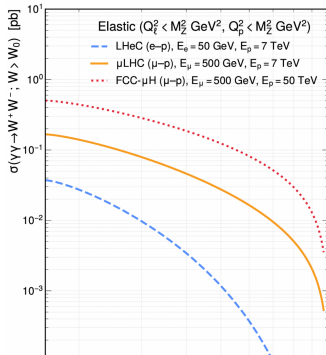
Updated paper benchmark

The latest draft replaces the previous illustrative point by

$$\Delta a_\tau = 5.0 \times 10^{-4}.$$

- Shown: updated **LHeC** elastic reference calculation from the paper draft.
- The EFT/SM ratio exposes the growing shape distortion in the hard $M_{\tau\tau}$ region.
- The same SMEFT fit should be repeated for $\text{LH}\mu\text{C}$, whose harder spectrum gives more leverage at high mass.

$\gamma\gamma \rightarrow W^+W^-$: a clean electroweak laboratory



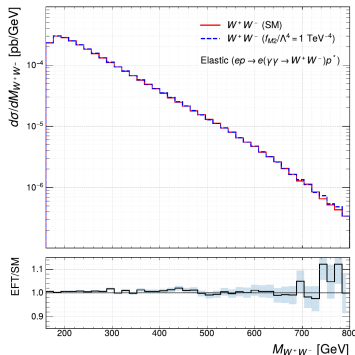
Elastic rates

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.037 \text{ pb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.166 \text{ pb}$$

- The W threshold lies inside the high-luminosity $\gamma\gamma$ regime.
- The process directly tests the SM $\gamma\gamma WW$ gauge structure.
- The high- M_{WW} tail is a clean lever arm for dimension-8 quartic-gauge operators.

Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow W^+W^-$



Dimension-8 EFT

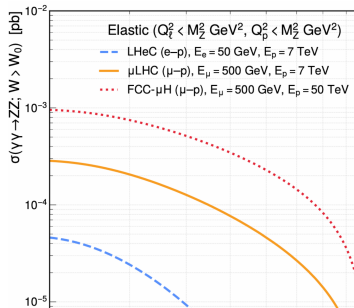
$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i}{\Lambda^4} \mathcal{O}_i,$$

$$\frac{f_{M2}}{\Lambda^4} = 1 \text{ TeV}^{-4}.$$

- Shown: updated **LHeC** elastic reference from the draft.
- EFT effects are tiny at low M_{WW} and grow in the hard tail.
- $\text{LH}\mu\text{C}$ supplies the harder $W_{\gamma\gamma}$ spectrum needed to extend this strategy quantitatively.

Scope: no detector-level $\text{LH}\mu\text{C}$ exclusion is claimed here; this is a defined next **HELP4CERN** study.

$\gamma\gamma \rightarrow ZZ$: a loop-induced Standard-Model channel



- In the SM, $\gamma\gamma \rightarrow ZZ$ is absent at tree level and proceeds through loops.
- The small SM rate makes the channel a clean window on anomalous neutral quartic interactions.
- LH μ C extends the accessible high- M_{ZZ} regime.

Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.046 \text{ fb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.284 \text{ fb}$$

Neutral quartic gauge couplings in $\gamma\gamma \rightarrow ZZ$

Rate model used in the updated study

$$\sigma_{\text{tot}} \simeq \sigma_{\text{SM}}^{\text{loop}} + \sigma_{\text{AQGC}}^{\text{tree}}.$$

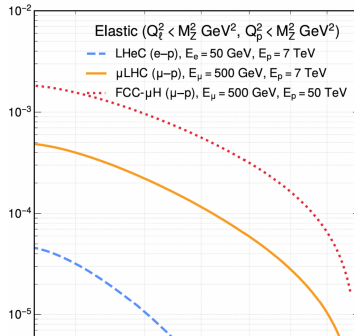
LHeC tagged-elastic reference

$\sigma_{\text{SM}} = 3.340 \times 10^{-5}$ pb; raw events assume 1 ab^{-1} before Z decays and detector effects.

f_{M2}/Λ^4	σ_{AQGC} [pb]	events	enhancement
0.1	1.17×10^{-8}	0.012	1.0004
1.0	1.17×10^{-6}	1.17	1.035
5.0	2.93×10^{-5}	29.3	1.88

- SM is loop induced; the quoted AQGC contribution is tree level.
- The table shows how an energy-growing neutral interaction can become competitive with a rare SM channel.
- These are **LHeC reference numbers**, not $\text{LH}\mu\text{C}$ projected limits; the harder $\text{LH}\mu\text{C}$ spectrum motivates the direct extension.

Heavy final states: $\gamma\gamma \rightarrow t\bar{t}$



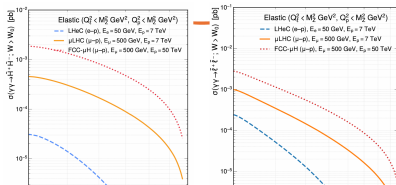
Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.046 \text{ fb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.486 \text{ fb}$$

- This is exclusive $\gamma\gamma \rightarrow t\bar{t}$ — **not** the much larger $\gamma g \rightarrow t\bar{t}$ photoproduction channel.
- The $2m_t$ threshold probes the hard end of the photon spectrum.
- The channel accesses the top electromagnetic interaction and high-energy new physics.

Beyond the Standard Model: charged-particle pair production in $\gamma\gamma$ collisions



Illustrative

100 GeV scalar (slepton) and fermion (higgsino) reference curves.

Representative channels:

$$\gamma\gamma \rightarrow \tilde{l}^+ \tilde{l}^-, \quad \gamma\gamma \rightarrow \tilde{\chi}^+ \tilde{\chi}^-.$$

- Rates are governed mainly by electric charge and mass, complementary to strongly interacting searches.
- Forward kinematic closure is attractive for compressed spectra with soft visible products and missing momentum.

Updated interpretation

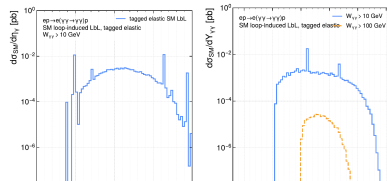
The draft notes that 100 GeV is now illustrative. Current constraints motivate roughly 300 GeV sleptons and 250–300 GeV higgsino benchmarks; LHeC rates near those thresholds are tiny, making a dedicated LH μ C re-evaluation especially relevant.

Light-by-light scattering, Higgs production, and other $\gamma\gamma$ opportunities



Mature calculation: $\gamma\gamma \rightarrow \gamma\gamma$

SM light-by-light scattering is loop induced; all charged SM particles contribute to the box amplitude.

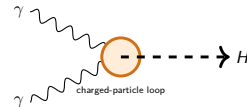


Updated

LHeC tagged-elastic SM pseudorapidity distributions.

Exploratory: $\gamma\gamma \rightarrow H$

The Higgs channel is loop induced and rare, but it probes the effective $H\gamma\gamma$ interaction in an exceptionally clean initial state.



LH μ C opportunity

The harder photon spectrum extends the high-mass LbL region and other loop-induced or resonant searches.

What does two-photon physics require from the $\text{LH}\mu\text{C}$ experiment?



- **Wide central tracking and lepton acceptance**, including robust τ reconstruction.
- **Efficient $W/Z/t$ reconstruction** for heavy $\gamma\gamma$ final states.
- **Exclusivity and acoplanarity** observables with sensitivity to low extra activity.
- **Missing-momentum performance** for neutrinos and compressed BSM signatures.
- **Scattered-muon measurement** to constrain the lepton-side photon kinematics.
- **Forward proton tagging where feasible**, together with control of proton dissociation.
- Exploit the defining advantage: the **clean, low-pile-up μp environment**.

Figure placeholder

Suggested figure: $\text{LH}\mu\text{C}$ interaction-region schematic with central detector, scattered-muon acceptance and a near-beam proton tag.

Physics–detector connection

The $\gamma\gamma$ programme links central reconstruction, lepton tagging and near-beam instrumentation directly to the usable photon luminosity and kinematic closure.

Summary and outlook



- LH μ C turns μp collisions into a **high-energy $\gamma\gamma$ laboratory** with a substantially harder photon spectrum than the LHeC benchmark.
- EPA and $S_{\gamma\gamma}$ connect parent μp luminosity transparently to elastic and proton-dissociative two-photon subprocesses.
- $\tau^+\tau^-$ and W^+W^- provide high-statistics precision/EFT channels; sensitivity lives in their hard invariant-mass tails.
- Loop-induced ZZ , exclusive $t\bar{t}$, light-by-light scattering, H , and charged BSM pairs extend the programme to rare quantum processes and heavy thresholds.

Bottom line: high-energy $\gamma\gamma$ physics is one of the distinctive physics opportunities of LH μ C.



The HELP4CERN collaboration



HELP4CERN collaboration focuses on the physics case for high-energy, high-luminosity lepton-hadron colliders at CERN.

- assess the physics potential of future lepton-hadron collider options,
- develop benchmark studies for Higgs, electroweak, top, QCD, and BSM physics,
- provide a common framework for phenomenology, detector studies, and future projections,
- new contributions and physics studies are very welcome,

Contact: help-4-cern@cern.ch

Visit: <https://help4cern.web.cern.ch>

White Paper on High Energy, High Luminosity Lepton-Hadron Colliders at CERN



D. Acosta^a D. Britzger^b A.C. Canbay^c K. Cheung^{d,e,f} B. Dagli^g L. Forthomme^h
 B. Holzerⁱ U. Kaya^j H. Khanpour^h P. Kotko^h M. Kumar^h B. Mellado^h S. Morettiⁱ
 M. Otani^{im} K. Piotrkowski^h D. Schulteⁱ A.M. Staśto^h S. Sultansoy^g I. Tapan^o
 S. Tentori^p M. Topp-Mugglestone^q Z.S. Wang^r Y. Yamazaki^s C. Zhang^t G. Ziarko^h
 F. Zimmermannⁱ

Next steps for HELP4CERN:

- turn the current benchmark studies into a coherent white-paper assessment of high-energy lepton-hadron options at CERN by the end of 2026;
- foster a worldwide study effort, in close coordination with the LHeC Study Group, aimed at preparing a dedicated **HELP4CERN proposal** by 2028.